

STAWELL, Australia

WMO: 948360

Lat: 37.0719S

Lon: 142.7400E

Elev: 236

StdP: 98.53

Time Zone: 10.00 (AUS)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.8	2.0	-4.2	2.7	25.7	-2.1	3.3	21.3	9.2	11.4	8.4	11.3	1.1	110	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.6	36.0	17.4	33.6	17.0	31.4	16.6	20.2	27.3	19.2	27.6	18.2	27.3	5.2	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.7	21.8	16.7	12.2	21.1	15.3	11.2	20.3	59.1	26.6	55.5	27.1	52.3	27.3	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	8.2	7.4	-1.1	40.9	0.8	1.9	-1.7	42.2	-2.2	43.3	-2.6	44.4	-3.2	45.8		
			-1.3	21.8	0.9	1.3	-1.9	22.7	-2.4	23.4	-2.9	24.1	-3.5	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.3	20.9	20.7	18.4	14.7	11.3	8.9	8.4	9.0	11.0	13.5	16.6	18.7
	DBStd	5.48	4.09	3.79	3.72	3.06	2.41	1.89	1.80	2.07	2.67	3.46	3.98	4.02
	HDD10.0	178	0	0	0	2	14	42	56	43	16	4	1	0
	HDD18.3	1797	17	17	45	117	219	282	309	288	220	156	82	44
	CDD10.0	1749	338	301	260	142	53	10	5	13	47	114	198	269
	CDD18.3	326	97	84	47	7	0	0	0	0	1	7	29	54
	CDH23.3	4552	1364	1007	584	91	1	0	0	0	8	144	464	888
	CDH26.7	2080	704	461	235	18	0	0	0	0	1	43	188	431
Wind	WSAvg	3.5	4.2	4.0	3.8	3.1	2.8	2.8	3.0	3.2	3.3	3.6	3.8	4.0
Precipitation	PrecAvg	565	36	28	33	41	57	60	67	65	64	58	35	25
	PrecMax	880	90	190	116	168	131	168	148	130	142	185	86	74
	PrecMin	223	2	0	1	3	6	8	11	11	15	8	7	0
	PrecStd	145	24	37	30	38	33	42	32	31	36	40	21	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	37.2	35.4	29.1	22.0	16.3	15.2	19.3	24.3	30.8	34.5	38.5
		MCWB	18.5	18.8	17.3	15.4	12.8	11.1	10.8	12.1	13.4	15.0	16.6	17.0
	2%	DB	36.7	34.3	31.9	25.9	19.2	14.5	13.6	16.5	21.2	27.3	31.5	34.7
		MCWB	17.9	17.7	16.9	14.4	12.5	10.7	9.9	10.6	13.0	13.7	15.9	16.7
	5%	DB	33.4	32.0	29.2	23.6	17.5	13.5	12.6	14.7	18.6	24.2	28.6	31.5
		MCWB	17.0	17.4	16.2	13.8	12.2	10.4	9.5	10.1	12.0	13.3	15.5	15.9
	10%	DB	30.6	29.4	26.5	21.4	15.8	12.6	11.8	13.2	16.5	21.2	25.6	28.4
		MCWB	16.6	16.9	15.7	13.2	11.7	10.0	9.1	9.5	11.3	12.6	14.7	15.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.6	21.8	19.7	17.0	15.1	13.0	12.1	13.1	15.7	17.2	19.3	20.6
		MCDB	26.9	28.9	26.3	23.2	16.8	13.9	13.4	16.5	20.6	23.1	28.0	26.0
	2%	WB	20.1	20.4	18.4	15.9	13.7	11.7	11.0	11.6	13.9	15.5	18.0	18.6
		MCDB	28.6	26.9	26.4	21.1	16.8	13.3	12.5	15.0	18.6	21.9	25.8	26.8
	5%	WB	19.0	19.2	17.5	15.0	13.0	11.0	10.2	10.7	12.7	14.3	16.9	17.5
		MCDB	29.4	26.8	25.8	21.1	16.2	12.7	11.7	13.7	17.6	21.4	24.0	27.3
	10%	WB	17.8	18.1	16.5	14.1	12.2	10.3	9.5	9.9	11.6	13.2	15.6	16.4
		MCDB	28.3	26.8	24.8	20.2	15.4	12.1	11.2	12.7	16.0	20.4	23.5	26.6
Mean Daily Temperature Range	5% DB	MDBR	14.6	13.6	12.4	11.1	8.7	7.2	6.9	8.1	9.7	11.7	12.9	14.0
		MCDBR	20.1	18.2	17.3	15.6	11.5	8.8	8.1	11.0	13.3	17.6	18.9	20.3
		MCWBR	5.7	5.3	5.7	6.5	6.1	5.6	5.1	6.2	6.5	6.7	6.0	5.9
	5% WB	MCDBR	15.3	13.4	13.4	11.9	9.2	7.0	6.7	9.4	11.5	13.3	14.1	14.5
		MCWBR	5.9	5.2	5.3	5.7	5.4	5.0	4.7	5.7	6.2	6.4	5.5	5.8
Clear-Sky Solar Irradiance	taub	0.340	0.331	0.315	0.306	0.299	0.290	0.288	0.302	0.324	0.328	0.333	0.327	
	taud	2.486	2.533	2.570	2.575	2.572	2.592	2.598	2.536	2.464	2.466	2.486	2.525	
	Ebn at Noon	993	980	957	907	853	834	859	897	931	971	994	1011	
	Edh at Noon	115	105	93	81	71	65	68	83	102	111	114	111	
All-Sky Solar Radiation	RadAvg	7.54	6.65	5.25	3.69	2.47	1.96	2.13	3.02	4.20	5.58	6.59	7.37	
	RadStd	0.42	0.32	0.26	0.21	0.12	0.16	0.13	0.19	0.29	0.44	0.43	0.38	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	+68
(47)	Regional (34 neighbors)	+0.47	N/A	N/A	+0.91	N/A	N/A	N/A	-127	+164	+52

Nomenclature: See separate page